

Personal Learning Journey

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Customer Segmentation Model Comparison and Validation

Project context. Our group analysed the UCI Online Retail dataset to identify customer segments and segment-specific product associations for retention, reactivation and cross-selling decisions. This reflection explains my individual contribution, learning and development during the project.

My contribution

My main responsibility was to construct the customer-segmentation baseline, compare clustering methods and select a defensible final solution. I created a quartile-based RFM scoring baseline and evaluated K-Means, Agglomerative Clustering, Gaussian Mixture Models and DBSCAN. The experiment design compared core and enhanced features, raw and log-transformed representations, StandardScaler and RobustScaler, and different cluster or parameter choices. In total, 192 unique clustering experiments were assessed using Silhouette, Davies-Bouldin, Calinski-Harabasz, cluster-size balance and DBSCAN noise. I then profiled the final clusters, assigned business-readable names and prepared one validated segment assignment for each of the 4,334 customers.

What I learned

The strongest lesson was that the best numerical score is not always the best business model. Some raw-feature solutions produced very high silhouette values by isolating extreme customers; the highest-scoring raw solution placed approximately 99.82% of customers in one cluster. Although this looked impressive statistically, it could not support useful marketing decisions. I learned to interpret metrics together, inspect cluster sizes and return to original-scale customer profiles before choosing a model. Comparing the RFM baseline with data-driven clusters also showed that a transparent baseline is valuable even when it is not selected.

Challenge and judgement

Selecting the number of clusters required a real trade-off. The three-cluster candidate had a slightly better silhouette and stronger stability, while five clusters provided clearer lifecycle detail. I selected K-Means with five clusters using log-transformed RFM features and StandardScaler because it kept competitive metrics, produced cluster shares from 8.26% to 26.21%, and supported distinct actions. Its silhouette was 0.319 and Davies-Bouldin index was 0.983. I tested ten random seeds and thirty repeated 80% customer samples; mean ARI values of 0.985 and 0.931 showed strong reproducibility. This process improved my critical judgement about unsupervised validation.

Teamwork and future development

My work depended on reliable feature definitions, and the association-mining stage depended on my customer assignments. I therefore documented configuration choices, exported validated identifiers and reviewed segment profiles with the team before naming them Hibernating Low Value, Recent Low Engagement, At Risk Regular, Loyal High Value and Champions. AI tools supported debugging and explanation, but I checked every recommendation against stored metrics, plots and stability results. In future, I would test out-of-time stability, soft membership and campaign outcomes. I learned to present clustering as decision support rather than claiming that the algorithm discovered permanent or objectively true customer types.